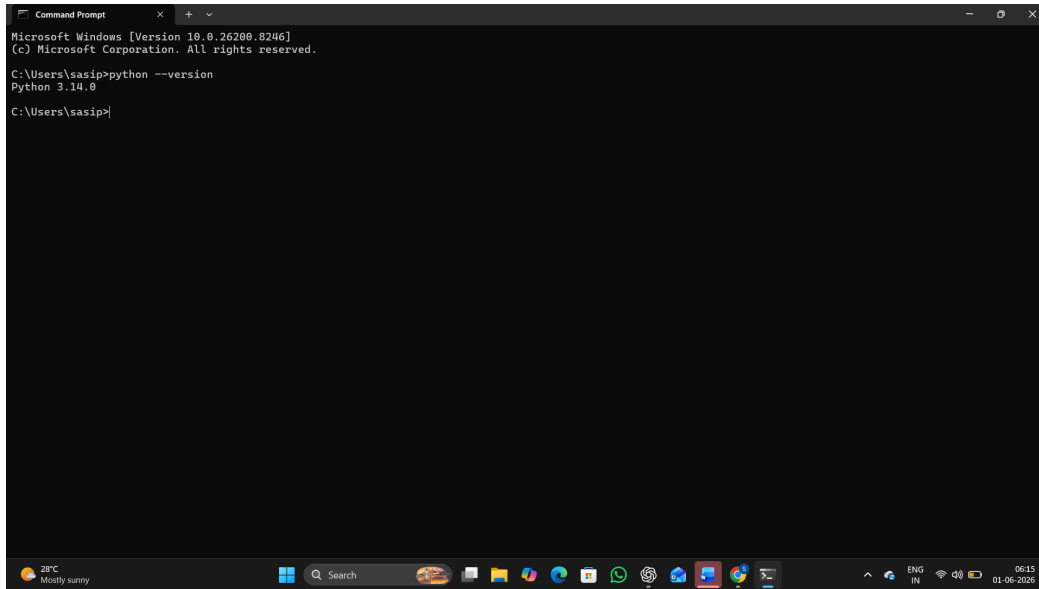


DAY-1 TASK REPORT

Task to complete:-Installations

(1) Python installation completed



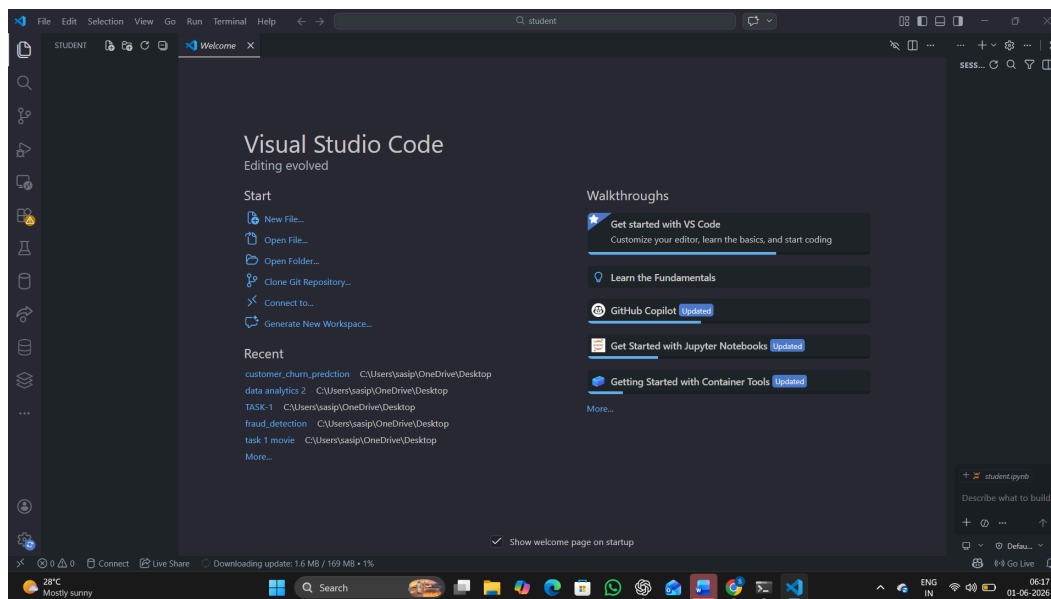
```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sasip>python --version
Python 3.14.0

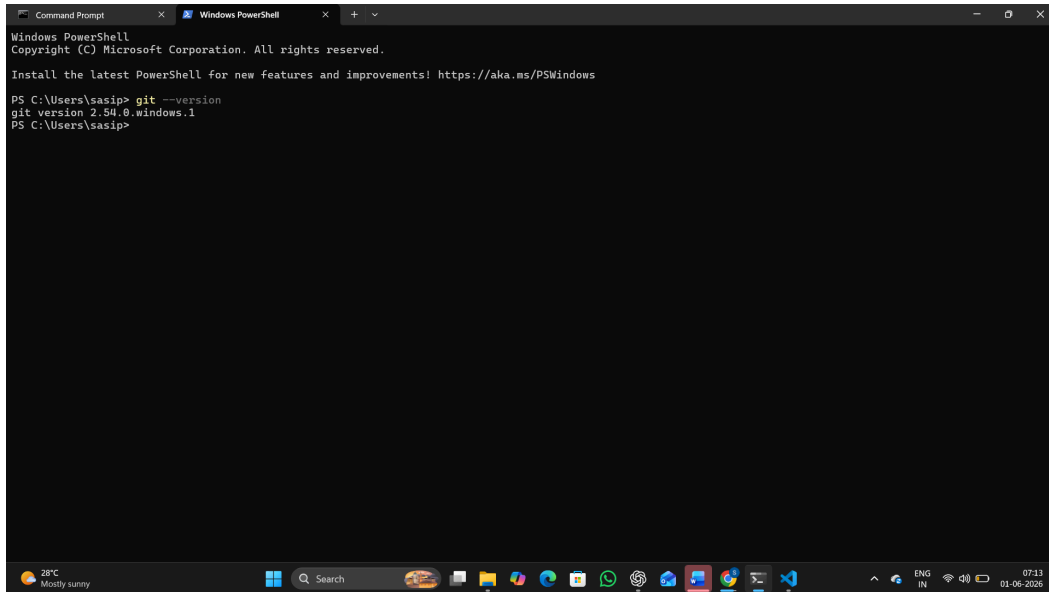
C:\Users\sasip>
```

The screenshot shows a Windows Command Prompt window. The title bar reads "Command Prompt". The window content displays the command prompt output for the command `python --version`, which returns `Python 3.14.0`. The user's current directory is `C:\Users\sasip>`. The Windows taskbar at the bottom shows the date and time as 06:15 on 01-06-2026, and the weather as 28°C Mostly sunny.

(2) vs code installation completed



(3) Git installation completed



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\sasip> git --version
git version 2.54.8.windows.1
PS C:\Users\sasip>
```

Create & Setup GitHub Account :-

Objective:

To create a GitHub account, log in successfully ,and configure the GitHub profile for software development and project management.

Requirements:

- (1) Internet connection
- (2) Unique Email address
- (3) chrome

Steps involved in GitHub account creation:

Step 1 : Create a GitHub Account

1. Open the GitHub website in chrome.
2. Click on the SIGN UP button and this SIGN IN use only first time of creating account from next we use SIGN IN .
3. Enter a valid email address which is unique.
4. Create a strong password of our choice.
5. Choose a unique username it should our name .

6. Complete the verification process.
7. After completion of process click on CREATE ACCOUNT.

Step 2 : login to GitHub

1. Open the GitHub website.
2. Click SIGN IN button.
3. Enter the registered email address or username.
4. Click Sign In to access the GitHub dashboard.

Step 3 : Setup GitHub profile

1. click on the profile icon in the top-right corner .
2. then select your profile.
3. [To](#) upload our details in github click edit profile.
4. Fill details like upload photo, full name, write short bio about your skills and interests, etc.
5. after that save the changes.

Result:

Finally the github account was created successfully.

Research:

(1) what is SLM ?

- SLM stands for *small language model* is an AI model that understands and generates human language like a Large Language Model (LLM), but it has **fewer parameters**, requires **less memory**, and runs **faster**.

Example:

SLM: Smaller models designed for specific tasks such as chatbots, text classification, summarization, or mobile applications.

Features of SLM :

- 1.smaller size
 - use fewer parameters than LLMs.
 - Easier to store and deploy.
- 2.Faster performance
 - Generates responses more quickly.
 - Suitable for devices with limited resources.
- 3.Lower cost
 - Requires less computing power.
 - Cheaper to train and run.

(2)what is RAG system?

- **RAG** stands for **Retrieval-Augmented Generation**. It is an AI technique that combines **information retrieval** with **text generation** to provide more accurate and up-to-date answers.
- *Retrieve relevant information from documents, databases, or websites .*
- *Generate an answer using a language model based on the retrieved information.*

Components of a RAG System :

- 1.Knowledge Base
- 2.Retriever
- 3.Vector Database
- 4.Language model

Why use RAG:

Without RAG

- AI answers only from its training data.
- Information may be outdated.
- Can produce incorrect facts (hallucinations).

With RAG

- Uses current documents.
- More accurate responses.
- Can answer organization-specific questions

Real-World Example : College Chatbot

A college uploads:

- Admission rules
- Fee structure
- Course details

When a student asks:

"What is the fee for Computer Science?"

The RAG system:

1. Searches the college documents.
2. Finds the fee information.
3. Generates an answer based on the document.

Technologies Used in RAG

- Python
- Machine Learning
- Natural Language Processing (NLP)
- Vector Databases
- Embedding Models
- Large Language Models (LLMs)

(3) Difference between LLM and SLM ?

Feature	LLM (Large Language Model)	SLM (Small Language Model)
Full Form	Large Language Model	Small Language Model
Model Size	Very large	Small
Number of Parameters	Billions to trillions	Millions to a few billions
Speed	Slower	Faster
Memory Requirement	High	Low

Cost	Expensive	Less expensive
Accuracy	Higher	Moderate
Hardware Needed	Powerful servers/GPUs	Can run on laptops and mobile devices
Knowledge & Reasoning	Better	Limited compared to LLMs
Use Cases	Advanced chatbots, content generation, coding assistants	Mobile assistants, simple chatbots, text classification

“LLM (Large Language Model) is a large AI model with billions of parameters that provides highly accurate and intelligent responses but requires more computing resources.

SLM (Small Language Model) is a compact AI model with fewer parameters that runs faster and uses less memory but has lower capability compared to LLMs.”

RAG with LLM :

Question



RAG Retrieves Data



LLM Processes Data



Detailed Answer

Example

Question: Explain the company's leave policy.

LLM Output:

A detailed explanation including leave types, eligibility, and rules.

Benefits

- Better reasoning
- More detailed responses
- Higher accuracy

RAG with SLM

Question



RAG Retrieves Data



SLM Processes Data



Simple Answer

Example

Question: How many leave days are allowed?

SLM Output:

20 paid leave days per year.

Benefits

- Faster responses
- Less memory usage
- Lower cost
- Can run on laptops and mobile devices